Nikodym sets: a power saving from finite-grid interpolation

Research note

5 September 2026

Abstract

We prove

$$\mathrm{Nik}(d,q) \geq q^d - (8d^2+1)q^{d-2^{1-d}} \qquad (d \geq 2)$$

for every prime power q. The main observation compares interpolation on the whole finite grid with interpolation on an irreducible carrier. Removing the conditions at private points supplies a polynomial cut whose degree is independent of the degrees of the carrier's defining equations. A dimension induction then halves the saving at each step. The proof uses monomial counting, finite-grid Hermite interpolation, linear Noether normalization, and the elementary degree bound for a proper hypersurface section. It uses no resolution of singularities, intersection pairing, or restriction on the characteristic.

1 The statement for families with private points

Fix an algebraic closure K of $\mathbb{F}_q$. An affine $\mathbb{F}_q$-line means its extension to a line over K as well as its q rational points, as appropriate. A family $\mathcal{L}$ of distinct affine $\mathbb{F}_q$-lines has *private points* if for each $\ell \in \mathcal{L}$ there is a specified $b_\ell \in \ell(\mathbb{F}_q)$ lying on no other line of $\mathcal{L}$. In particular, the b_ℓ are distinct.

Theorem 1. *Let $d \geq 2$, $1 \leq k \leq d$, and let $A \subseteq \mathbb{P}^d_K$ be an integral projective variety of dimension k and degree Δ. Suppose A contains the projective closures of L affine $\mathbb{F}_q$-lines having private points. Then*

$$L \leq C_d \Delta q^{k-2^{1-k}}, \qquad C_d = 8d^2+1. \tag{1}$$

The variety A need not be defined over $\mathbb{F}_q$.

Corollary 2. *For every $d \geq 2$, every prime power q, and every Nikodym set $N \subseteq \mathbb{F}_q^d$,*

$$|\mathbb{F}_q^d \setminus N| \leq (8d^2+1)q^{d-2^{1-d}}. \tag{2}$$

Consequently,

$$\mathrm{Nik}(d,q) = q^d - O_d(q^{d-2^{1-d}}).$$

Deduction of the corollary. Put $B = \mathbb{F}_q^d \setminus N$. For each $b \in B$, choose a witness line ℓ_b with $B \cap \ell_b = \{b\}$. These lines are distinct, and b is a private point of ℓ_b. Apply Theorem 1 with $A = \mathbb{P}^d$, $k = d$, and $\Delta = 1$. $\square$

2 Two polynomial dimension estimates

For a projective variety $A \subseteq \mathbb{P}^d$, write $H_A(t)$ for its homogeneous Hilbert function: the dimension of degree-t forms modulo those vanishing on A. If A meets the standard affine chart, this is also the dimension of restrictions to $A \cap \mathbb{A}^d$ of affine polynomials of total degree at most t.

Lemma 3 (Normalized Hilbert functions)**.** *For any homogeneous ideal in $K[X_0, \ldots, X_d]$, the Hilbert function H of its quotient satisfies*

$$\frac{H(u)}{\binom{u+d}{d}} \leq \frac{H(t)}{\binom{t+d}{d}} \qquad (0 \leq t \leq u). \tag{3}$$

Proof. Replace the ideal by an initial monomial ideal, which preserves its Hilbert function. Call a monomial *standard* if it is outside this ideal. Every divisor of a standard monomial is standard.

Give the edge from X^α of degree t to $X^{\alpha+e_i}$ of degree $t+1$ weight $\alpha_i + 1$. Every standard child has only standard parents, and its total incoming weight is $t+1$. The total outgoing weight from any standard parent is $t+d+1$. Counting weighted edges therefore gives

$$(t+1)H(t+1) \leq (t+d+1)H(t).$$

This is the adjacent-degree case of (3); iterate. □

Lemma 4 (Degree upper bound)**.** *If $A \subseteq \mathbb{P}^d_K$ is integral of dimension k and degree Δ, then*

$$H_A(t) \leq \Delta \binom{t+k}{k} \qquad (t \geq 0). \tag{4}$$

Proof. Here is a proof of this standard degree estimate. Choose a finite linear projection $\pi : A \to \mathbb{P}^k$ of degree Δ whose function field extension is separable. Such a projection exists over the infinite perfect field K: linear Noether normalization gives finiteness, and generic linear coordinates are separating coordinates on the smooth locus.

Choose an affine base point with Δ distinct unramified preimages. The completed local rings at these preimages have the same k base coordinates. Taking the Taylor coefficients of total degree at most t at all these branches defines a linear map from degree-t polynomial restrictions into a vector space of dimension $\Delta\binom{t+k}{k}$.

This map is injective. Indeed, if a restriction P has all these coefficients zero, its field norm to the base vanishes to order at least $\Delta(t+1)$. On the other hand, that norm is a polynomial of degree at most Δt. To justify the degree statement, the homogeneous coordinate ring of A is finite over the polynomial ring in the $k+1$ projection coordinates. The norm of a degree-t homogeneous element lies in that polynomial ring by integrality and its integral closedness, and has homogeneous degree Δt. Dehomogenize on the chosen affine chart.

A nonzero polynomial cannot vanish at a point to order greater than its total degree. Thus the norm is zero, which forces $P = 0$ because a nonzero field element has nonzero norm. The Taylor map is injective, proving the bound. All Taylor coefficients are formal coefficients; no division by factorials is involved. □

3 Interpolation on the entire finite grid

For $x \in A \cap \mathbb{A}^d(K)$ and $r \geq 1$, put

$$j_{A,x}(r) = \dim_K(\mathcal{O}_{A,x}/\mathfrak{m}^r_{A,x}).$$

Set $j_{A,x}(r) = 0$ when $x \notin A$.

Lemma 5 (Local minimum)**.** *For every point x of an integral k-dimensional variety,*

$$j_{A,x}(r) \geq \binom{r+k-1}{k}. \tag{5}$$

Proof. The associated graded ring $\mathrm{gr}_{\mathfrak{m}}\,\mathcal{O}_{A,x}$ is a standard graded K-algebra of dimension k. Linear Noether normalization gives k algebraically independent degree-one elements in this ring. Their monomials of degree less than r are linearly independent. There are $\binom{r+k-1}{k}$ such monomials, and the sum of the first r graded dimensions is $j_{A,x}(r)$. □

Lemma 6 (Grid interpolation)**.** *For every integer $r \geq 1$, put*

$$U = q(r-1) + d(q-1). \tag{6}$$

Restriction of degree-at-most-U polynomials on $A \cap \mathbb{A}^d$ surjects onto

$$\bigoplus_{x \in \mathbb{F}_q^d \cap A} \mathcal{O}_{A,x}/\mathfrak{m}_{A,x}^r.$$

In particular,

$$\sum_{x \in \mathbb{F}_q^d} j_{A,x}(r) \leq H_A(U). \tag{7}$$

Proof. In $R = K[X_1, \ldots, X_d]$, let

$$Z_i = X_i^q - X_i, \qquad J = (Z_1, \ldots, Z_d).$$

The polynomials $X_i^q - X_i$ have distinct roots, exactly $\mathbb{F}_q$. Thus $J = \bigcap_{x \in \mathbb{F}_q^d} \mathfrak{m}_x$. The distinct maximal ideals are pairwise comaximal, so

$$J^r = \bigcap_{x \in \mathbb{F}_q^d} \mathfrak{m}_x^r, \qquad R/J^r \cong \bigoplus_{x \in \mathbb{F}_q^d} R/\mathfrak{m}_x^r.$$

Repeated use of $X_i^q = Z_i + X_i$ shows that R is spanned over $K[Z_1, \ldots, Z_d]$ by the monomials X^α with $0 \leq \alpha_i < q$. Hence R/J^r is spanned by

$$X^\alpha Z^\gamma, \qquad 0 \leq \alpha_i < q, \quad |\gamma| < r.$$

Every such polynomial has degree at most $d(q-1) + q(r-1) = U$. Consequently ambient polynomials of degree at most U interpolate arbitrary ambient jets of order less than r at every grid point simultaneously.

Now quotient each jet space by the affine ideal of A. At a point $x \in A$ the resulting quotient is precisely $\mathcal{O}_{A,x}/\mathfrak{m}_{A,x}^r$; at a point outside A it is zero. The map factors through polynomial restrictions on A. This proves both surjectivity and (7). □

One useful consequence, requiring no separate point-counting theorem, is

$$|A \cap \mathbb{F}_q^d| \leq \Delta q^k. \tag{8}$$

Indeed, Lemmas 5, 6, and 4 give

$$|A \cap \mathbb{F}_q^d| \binom{r+k-1}{k} \leq \Delta \binom{q(r-1) + d(q-1) + k}{k}.$$

Divide by the binomial on the left and let the integer r tend to infinity, keeping q fixed.

4 A polynomial cut independent of the defining degrees

Proposition 7. *Suppose A and its $L > 0$ lines satisfy the hypotheses of Theorem 1. Define*

$$\rho = \frac{L}{\Delta q^k}, \qquad a = 8d^2, \qquad r = \left\lceil \frac{a}{\rho} \right\rceil.$$

If $r \le q$, there is a polynomial G of degree at most

$$T = r(q-1) - 1 \tag{9}$$

which does not vanish identically on A and vanishes identically on every selected line.

Proof. Let B_0 be the set of the L private points. Impose on polynomial restrictions on A the linear conditions

$$G \in \mathfrak{m}_{A,x}^r \qquad (x \in (A \cap \mathbb{F}_q^d) \setminus B_0).$$

The number of conditions is at most

$$\begin{aligned} \sum_{x \in \mathbb{F}_q^d \setminus B_0} j_{A,x}(r) &= \sum_{x \in \mathbb{F}_q^d} j_{A,x}(r) - \sum_{b \in B_0} j_{A,b}(r) \\ &\le H_A(U) - L\binom{r+k-1}{k}, \end{aligned} \tag{10}$$

where U is given by (6).

We show that the last expression is strictly less than $H_A(T)$. By (8), $0 < \rho \le 1$, so $r \ge 8d^2$. Writing $s = U - T$, we have

$$s = r + (d-1)q - d + 1 \le dq, \qquad T + 1 = r(q-1) \ge rq/2.$$

The normalized Hilbert inequality gives

$$H_A(U) - H_A(T) \le H_A(T)\left(\frac{\binom{U+d}{d}}{\binom{T+d}{d}} - 1\right).$$

Moreover,

$$\frac{\binom{U+d}{d}}{\binom{T+d}{d}} = \prod_{i=1}^{d}\left(1 + \frac{s}{T+i}\right) \le \exp\left(\frac{ds}{T+1}\right) \le \exp\left(\frac{2d^2}{r}\right).$$

Since $2d^2/r \le 1/4$, the inequality $e^z - 1 \le 2z$ for $0 \le z \le 1/4$ shows that the parenthesis is at most $4d^2/r$. Finally, for $1 \le i \le k$,

$$T + i = r(q-1) + i - 1 \le q(r + i - 1),$$

and therefore

$$\binom{T+k}{k} \le q^k \binom{r+k-1}{k}.$$

Combining these estimates with Lemma 4 yields

$$\begin{aligned} H_A(U) - H_A(T) &\le \frac{4d^2}{r}\Delta q^k \binom{r+k-1}{k} \\ &\le \frac{\rho}{2}\Delta q^k \binom{r+k-1}{k} = \frac{L}{2}\binom{r+k-1}{k}. \end{aligned} \tag{11}$$

Equations (10) and (11) prove the strict dimension inequality. There is consequently a nonzero restriction on A of a polynomial G of degree at most T satisfying all the imposed conditions.

Each selected line contains exactly one point of B_0, its own private point. At each of its other $q-1$ rational points, restriction from A to the line carries $\mathfrak{m}_{A,x}^r$ into the rth power of the line's maximal ideal. Thus G restricted to the line has $q-1$ roots of multiplicity at least r. Its degree is at most $T < r(q-1)$, so it vanishes identically on the line. $\square$

5 Dimension induction and the exponent

We recall the elementary degree fact used here. If an integral k-dimensional projective variety of degree Δ is not contained in a degree-e hypersurface, its intersection has pure dimension $k-1$, and the sum of the degrees of its reduced irreducible components is at most $e\Delta$. The degree statement follows by taking the Hilbert function difference $H_A(t) - H_A(t-e)$; its leading term is $e\Delta t^{k-1}/(k-1)!$. The principal ideal theorem and the dimension formula give the dimension assertion. Intersection multiplicities are positive, so discarding them can only decrease the degree sum.

Proof of Theorem 1. Fix d, put $a = 8d^2$ and $C = a+1$, and induct on k. If $k = 1$, an integral curve containing a line is that line. Thus $L \le \Delta$, which is stronger than the asserted bound. We can also assume $L > 0$ throughout.

Suppose $k \ge 2$ and the result holds in dimension $k-1$. Write $\rho = L/(\Delta q^k)$; by (8), $0 < \rho \le 1$. If $\rho < C/q$, then

$$L < C\Delta q^{k-1} \le C\Delta q^{k-2^{1-k}},$$

as required. Otherwise put $r = \lceil a/\rho \rceil$. Then

$$r \le \frac{a}{\rho} + 1 \le \frac{C}{\rho} \le q.$$

Proposition 7 gives a proper polynomial cut of degree at most $T = r(q-1) - 1 < rq$ containing all L lines.

Let $A_1, \dots, A_m$ be the reduced irreducible components of this cut. Each has dimension $k-1$, and

$$\sum_{i=1}^{m} \deg A_i \le \Delta T.$$

Assign each line to one component that contains it. Every assigned subfamily retains its private points. Put $\alpha = 2^{2-k}$, the saving in dimension $k-1$. Applying the inductive bound on each component and summing gives

$$L \le C \sum_i (\deg A_i) q^{k-1-\alpha} \le C\Delta T q^{k-1-\alpha} \le C\Delta r q^{k-\alpha}.$$

After division by Δq^k and use of $r \le C/\rho$,

$$\rho \le C r q^{-\alpha} \le \frac{C^2}{\rho} q^{-\alpha}.$$

Hence

$$\rho \le C q^{-\alpha/2} = C q^{-2^{1-k}},$$

which completes the induction. □

Remark 8 (Constants and lower-order terms). The constant was chosen for a short uniform proof and is not optimized. For the Nikodym problem the conclusion can also be written

$$q^d - |N| \le q^{d-2^{1-d}+o(1)}.$$

More explicitly, the $o(1)$ in this display may be taken to be $\log(8d^2+1)/\log q$. The first exponents are

$$\varepsilon_2 = \frac{1}{2}, \quad \varepsilon_3 = \frac{1}{4}, \quad \varepsilon_4 = \frac{1}{8}, \quad \varepsilon_5 = \frac{1}{16}.$$

No step uses ordinary differentiation, so there is no large-characteristic hypothesis.